Feature Flag Value			Value of OccupancySensorTypeBitmap	Value of OccupancySensorType
1	1	1	PhysicalContact + PIR + Ultrasonic	PIRAndUltrasonic **

[*] In case the feature flags PIR, US and PHY are all set to 0, these attributes are set to mimic a PIR sensor since this likely will give the best compatibility with clients implementing a cluster revision <= 4, which are not aware of the feature flags and the added modalities. In this case, timing parameters (when HoldTime is supported) are expressed using the PIROccupiedToUnoccupiedDelay, PIRUnoccupiedToOccupiedDelay and PIRUnoccupiedToOccupiedThreshold attributes.

[**] These combinations of sensor type were not defined for OccupancySensorType in ClusterRevision <= 4, so they are mapped to a combination which *was* defined in that version.

2.7.6.3. HoldTime Attribute

This attribute SHALL specify the time delay, in seconds, before the sensor changes to its unoccupied state after the last detection of occupancy in the sensed area. This is equivalent to the legacy *OccupiedToUnoccupiedDelay attributes.

Low values of HoldTime SHOULD be avoided since they could lead to many reporting messages. A value 0 for HoldTime SHALL NOT be used.

The figure below illustrates this with an example of how this attribute is used for a PIR sensor. It uses threshold detection to generate an "internal detection" signal, which needs post-processing to become usable for transmission (traffic shaping). The bit in the Occupancy attribute will be set to 1 when the internal detection signal goes high, and will stay at 1 for HoldTime after the (last) instance where the internal detection signal goes low.

The top half of the figure shows the case of a single trigger: the bit in the Occupancy attribute will be 1 for the duration of the PIR signal exceeding the threshold plus HoldTime. The bottom half of the figure shows the case of multiple triggers: the second trigger starts before the HoldTime of the first trigger has expired; this results in a single period of the bit in the Occupancy attribute being 1. The bit in the Occupancy attribute will be set to 1 from the start of the first period where the PIR signal exceeds the threshold until HoldTime after the last moment where the PIR exceeded the threshold.

